SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

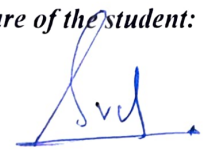
Annexure A

Descriptive Written Test

Code of Conduct:

I Saai Vardhan .D (192525205) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ARTIFICIAL INTELLIGENCE

CO1 Assessment - 1

CO1 - AT-3

Slot - B

CSA 1737

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1. A smart home automation system is beign designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type could be most effective for ensuring comfort, energy efficiency, and security.

Intelligent agent:-

An entity that perceives its environment through sensors and acts on it through actuators to achieve its goals. Here, the sensors are motion, temperature, and time/clock; the actuators are lights, AC/heater and door locks/alarm.

Types of agents and how each works in the Smart home

1. Simple Reflex Agent

- Uses only the current percept, with condition - action rules (no memory).
- Example :- If motion detected AND time = night → turn on ligh.
  If temperature >30° C → switch on Ac
- Limitation :- It can't learn preferences or handle partial Information (eg, it can't tell whether the person is still in the room).

2. Model-Based Reflex Agent

- Keeps an internal state, a model of how the world works and how its actions affect it.
- Example :- It remembers that someone entered the room even when there

is a motion (sleeping or reading), so it doesn't switch off the lights. It also tracks which doors and windows are open

· Limitation :- It knows the state but has no goal or sense of what is "best"

3. Goal - Based Agent

· Chooses actions that move it towards a goal, using search and planning.

· Example; Goal = "room at 24°C by 7PM" The agent plans when to pre-cool, or goal = "house Secured at night", so it locks the doors and arms the alarm.

· Limitation :- It doesn't compare how well the goal is met (comfort vs cost)

4. Utility - Based Agent

· Uses a utility function to choose the action with the highest "happiness" when

goals conflict.

- Example: Comfort and energy saving conflict. It weighs them, eg 25°C instead of 22°C saves 20% energy with little loss of comfort. It also weighs how much security is worth versus the false-alarm cost

- Learning improves it futher:- by observing the user's habits, it updates its utility function over time.

Agent	Memory	Goals	Handles trade-offs	Learns
Simple-reflex	No	No	No	No
Model-based	Yes	No	No	No
Goal-based	Yes	Yes	No	No
Utility-based	Yes	Yes	Yes	Can

Best approach:- a hybrid agent

- Simple reflex layer :- for instant, safety-critical responses (intruder detected → alarm; Smoke → alert)
- Model-based layer :- to track the home's state (who is home, which windows are open)
- Goal-based layer :- for planning (pre-heating, scheduling)
- Utility-based layer + learning to balance comfort, energy efficiency, and security and to adapt to user preferences.

No single agent can meet all three needs, Reflex rules give speed and reliability for security, while the utility-based and learning layers optimize comfort and energy cost. So a hybrid, learning utility-based agent is the most effective

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Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent types and search strategy would be most effective and justify your choice.

The robot is in an unknown maze with no prior knowledge, So only uniformed (blind) search is possible, Since no heuristic is available. The robot expands

States (positions) and finds a path to the exist.

1. Uniform Cost Search (UCS)

- Expands the node with the lowest path cost $g(n)$ first, using a priority queue.
- Example: If a step costs 1 for a straight move and 3 for turning or climbing, UCS finds the cheapest path, not just the fewest steps.
- Complete: Yes (if step costs $\geq \epsilon > 0$)
- Optimal → Yes
- Time and space $O(b^{(1 + \lfloor C^*/\epsilon \rfloor)})$, where C^* is the optimal cost.
- Advantage: Guarantees the least-cost path.
- Disadvantage: It explores in all directions and keeps all nodes in memory, so space grows exponentially.

2. Iterative Deepening Search (IDS)

- Repeats depth-limited DFS with limits 0, 1, 2 ... untill the goal is found. It combines the memory efficiency of DFS with the completeness of BFS.
  - Complete :- Yes (finite branching)
  - Optimal :- Yes (for unit step costs)
  - Time :- $O(b^d)$
  - Space :- $O(bd)$, which is linear
- Advantages :- Very low memory, which suits a robot with limited memory.
- Disadvantage :- It re-expands upper levels repeatedly (a small overhead, since most nodes are at the deepest level). It is not optimal when step costs differ.

3. Comparison

Feature	UCS	IDS
Strategy	Lowest cost first	Repeated depth-Limited DFS
Complete	Yes	Yes
Optimal	Yes (any positive cost)	Only for uniform cost
Time	$O(b^{(1+C^*/\epsilon)})$	$O(b^d)$
Space	Exponential	$O(bd)$, Linear.

4. Relating to intelligent agents.

Simple reflex :- Reacts only to the current percept (i.e. "wall ahead → turn"). It can loop forever in a maze because it has no memory

Model-based :- Remembers visited cells and builds a map, so it avoids loops. This is essential in an unknown maze.

Goal-based :- It's goal is "reach the exit" It ensures the search (UCS/IDS) to plan

a path.

. Utility- based :- It balances path length, energy use, and time when the paths differer in cost.

5. Best Combination

A model-based, goal-based agent using IDS (with UCS if step costs vary)

Justification :-

. The model (map memory) prevents loops and handles the partially observable maze.

. The goal-based design makes the robort plan toward the exit

. IDS needs only linear memory and guarantees finding a shortest path when step cost are equal

. An Utility based agent can be added when time and energy trade-offs matter.